

Worksheet 3: Perfect Competition

You may work together in groups to discuss and solve these problems, but each person is required to write and submit their own answers. AI use is not permitted. Answers are uploaded in PDF format to ELMS. Each question is worth 2 points. Partial credit may be given.

Part A

Suppose a competitive market consists of firms each with identical cost functions. Also assume this is a constant cost industry.

$$C(q) = 20 + q + 5q^2$$

1. Suppose initially, the market price is $p = 31$. How much output will each firm produce?
2. How much profit does each firm make at $p = 31$?
3. What is the shut-down price for this firm?
4. What is the break-even price for this firm?
5. Now, suppose the market demand is $Q_D = 602 - 2P$ and there are 20 firms in the market. What is the short-run market supply function? (Note: do not assume $p = 31$ anymore.) What is the market equilibrium price p^* in the short run?
6. Continuing with the scenario in 5, what is the output per firm q^* in the short-run equilibrium?
7. Continuing with the scenario in 5, what is the market output Q^* in the short-run equilibrium?
8. From the previous question, how much profit is each firm earning? Using that, do you expect firms to enter or exit the industry in the long run (increase/decrease/no change)? What will happen to the market price (increase/decrease/no change)? What will happen to the output per firm (increase/decrease/no change)? What will happen to the market output (increase/decrease/no change)?

Part B

Sandboxes are produced according to the following cost function:

$$c(q) = q^2 + 100$$

where the fixed cost of 100 represents an annual license fee the firms pay.

9. In the long run, what will be the equilibrium price?
10. The market demand for sandboxes is given by $Q_D = 1500 - 5p$. Find the long-run equilibrium market quantity.
11. Continuing with the long-run equilibrium, how many firms are operating?
12. Recent trends have increased the demand to $Q'_D = 2250 - 5p$. In the short run, what will be the new equilibrium price?